

Entropy of Identity in LLM-Mediated Environments: A Framework for Measuring and Stabilizing Algorithmic Identity Representation

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Abstract

Large Language Models increasingly determine how individuals are represented in digital environments. Unlike search engines that retrieve documents, LLMs reconstruct identity from incomplete, distributed data — introducing systematic distortion we term identity entropy: the divergence between a person's verifiable digital presence and their AI-generated representation. We formalize this divergence, introduce the Entity Life Cycle (ELC) framework as a practical stabilization architecture, and illustrate it through a documented case of an independent artist resolving competitive namespace interference without institutional resources. This work positions identity management as an engineering problem, not a visibility problem.

Keywords: identity entropy, LLM representation, entity disambiguation, AI-mediated identity, Entity Life Cycle, knowledge graphs, RAG systems, Native AI Branding, Delta of Intent

1. Introduction

When someone searches for a person today, they increasingly receive not a list of documents but a synthesized answer generated by a language model. This shift — from retrieval to reconstruction — has profound consequences for how identity is formed, maintained, and distorted in digital space.

For well-documented entities, reconstruction is stable. For emerging individuals — independent artists, researchers, entrepreneurs without institutional backing — the process is fragile. Sparse data, naming conflicts, and inconsistent sources cause generative models to compress identity into stereotypes, confuse entities with namesakes, or produce null results entirely.

This is not a visibility problem. It is an architectural one.

Modern LLMs increasingly operate via Retrieval-Augmented Generation (RAG), pulling external data into the context window at inference time. This means identity representation is shaped not only by training weights but by the structure and authority of crawlable artifacts. The Entity Life Cycle (ELC) framework addresses both layers simultaneously (Udalaya, 2026b). This paper also draws on the Native AI Branding methodology (Udalaya, 2026a) and the concept of Delta of Intent as a measure of human intervention in generative pipelines (Udalaya, 2026d).

This paper asks: how can individuals architect their digital presence to reduce systematic distortion in AI-mediated identity reconstruction — without relying on media coverage, Wikipedia inclusion, or institutional authority?

2. Identity Entropy: Formal Definition

2.1 Identity as a Distribution

Let $A = \{a_1, \dots, a_n\}$ be the set of verifiable digital artifacts associated with an individual: publications, persistent identifiers (ORCID, ISNI, MusicBrainz), knowledge graph nodes, structured metadata.

We define:

- $P_{\text{real}}(I)$ — distribution of identity signals from verified artifacts. Crucially, this is not the sum of all web mentions (which contain noise and errors) but a curated corpus of high-trust artifacts whose authority is established through persistent identifiers and cross-referencing.
- $P_{\text{model}}(I \mid M, X)$ — distribution of identity attributes reconstructed by model M given query X .

2.2 Identity Entropy

$$H_{\text{identity}} = D(P_{\text{real}}(I) \parallel P_{\text{model}}(I \mid M, X))$$

Where D is instantiated as Jensen-Shannon divergence (symmetric, bounded, empirically stable) or cosine distance in embedding space (practical approximation).

High identity entropy manifests as:

- Omission of distinctive attributes
- Compression into stereotypical categories
- High variance across repeated queries
- Confusion with namesakes (competitive namespace interference)

2.3 The Reconstruction Pipeline

LLM identity reconstruction operates in three stages:

1. Retrieval — aggregation from web corpora, knowledge graphs, social traces, internal priors; in RAG systems, live external retrieval is added at inference time
2. Compression $C: A \rightarrow Z$ — lossy, model-dependent; systematically discards low-frequency signals
3. Generation $G(Z, X) \rightarrow R_M(I)$ — reconstruction conditioned on query context

Identity entropy is introduced at the compression stage. Entities with sparse, inconsistent, or low-authority source data suffer disproportionate information loss.

3. Entity Life Cycle (ELC) Framework

3.1 Definition

$$ELC = \{S, M, V\}$$

- S — Structured self-declared data: authoritative first-person layer (biographies, methodology documents, JSON-LD schemas via schema.org/Person)
- M — Machine-readable metadata: persistent identifiers (ORCID, ISNI, MusicBrainz, Discogs), Wikidata nodes, Open Graph tags
- V — Verifiable external references: DOI-registered publications, press releases on indexed domains, third-party database entries

3.2 Mechanism

ELC does not control model output. It constrains the reconstruction space:

$P_model(I | M, X, ELC)$ converges toward $P_real(I)$ as $|ELC|$ increases

Structured, cross-referenced artifacts on high-authority crawlable domains progressively reduce the probability of misidentification, stereotype compression, and identity drift — across both parametric knowledge (training weights) and RAG retrieval. The framework was first formalized in Udalaya (2026b).

3.3 Implementation Protocol

For practitioners, ELC deployment follows four steps:

1. Identify — register persistent identifiers: ORCID (researchers), ISNI (creators), MusicBrainz/Discogs (musicians)
2. Structure — create JSON-LD markup linking name to identifiers via schema.org/Person
3. Publish — place anchor content (methodology, biography, work descriptions) on high-authority domains with permanent DOI links
4. Synchronize — align all public profiles to consistent syntax: full name, role, unique identifiers

3.4 Relation to Existing Frameworks

Self-Sovereign Identity (SSI) addresses authentication and credential verification but does not model semantic reconstruction by generative systems. Knowledge graphs support deterministic retrieval but assume static entity records, not probabilistic generative reconstruction. Entity resolution in NLP handles disambiguation across datasets but does not account for identity drift under generative synthesis. ELC extends all three by treating identity as a probabilistic reconstruction problem and positioning the individual as an active architect of their own retrieval input. It is designed to operate together with Native AI Branding (Udalaya, 2026a) and the Delta of Intent measure (Udalaya, 2026d).

4. Case Study: Resolving Competitive Namespace Interference

4.1 Context and Baseline

The subject is an independent music artist and marketing strategist operating under a stage name that shares its primary token with a globally prominent pop artist with tens of millions of monthly listeners. This creates acute competitive namespace compression: LLM training data is overwhelmingly dominated by the high-frequency entity, causing the low-frequency entity to be systematically excluded from reconstruction.

Pre-intervention state: queries for the stage name returned the competing artist exclusively. The compound query “[stage name] + Story” was interpreted as a reference to the high-profile artist’s social media content — a double misidentification: wrong entity, wrong semantic category. Queries for the full legal name returned null results. Maximum identity entropy.

4.2 ELC Intervention

Over approximately six months beginning early 2026, the subject deployed all three ELC layers:

- S layer: Bilingual (RU/EN) articles establishing two original methodologies — Native AI Branding and Entity Life Cycle Framework — published across Medium, Substack, Academia.edu, Hashnode, Dev.to, and a major Russian tech platform.
- M layer: ISNI (0000 0005 3033 4113), ORCID (0009-0004-9089-0859), MusicBrainz Artist ID, Discogs Artist ID, Zenodo DOIs for methodology documents, verified streaming platform artist profiles.
- V layer: Indexed press release, streaming metadata for debut single release (May 2026), structured AEO content blocks targeting LLM query clusters across multiple platforms.

4.3 Observed Outcome

Following intervention, LLM queries for the legal name and stage name began returning entity-specific responses: accurate description of the marketing strategist and Cinematic Pop artist, correct attribution of the debut release, and accurate representation of the original methodology. Namespace interference was resolved through query specificity: combined queries using full name or genre-qualified stage name yield unambiguous entity recognition across tested models. The progression from null/misidentified state to accurate cross-platform entity recognition occurred within approximately six months, without institutional support, media coverage, or financial expenditure on promotion.

Limitation: the absence of timestamped LLM query logs from the pre-intervention period means this case relies on qualitative observation. Future work should establish systematic logging protocols from the outset of any identity intervention.

5. Societal Implications

Three systemic consequences follow from this framework.

Representational inequality. Entities with resources to invest in structured data infrastructure will systematically achieve lower identity entropy. Those without — independent creators, researchers from under-resourced institutions, individuals from regions with sparse web coverage — face persistent misrepresentation in AI-mediated environments. This constitutes a new axis of digital inequality, compounding existing disparities in search visibility.

Institutional disintermediation. ELC demonstrates that individuals can reduce identity entropy without the traditional gatekeepers of identity authority: journalistic coverage, Wikipedia inclusion, or institutional databases. This is potentially democratizing — but also raises questions about the quality signals that replace institutional authority.

Entropy creep. Identity entropy is not static. As training data ages and new model versions are trained on different data distributions, entities must actively maintain their ELC infrastructure or risk gradual re-compression. Digital identity in the LLM era requires ongoing architectural maintenance, not one-time optimization.

6. Future Work

- Systematic measurement of identity entropy across LLM families using standardized query protocols and embedding-based divergence metrics
- Multi-subject longitudinal studies tracking entropy reduction over time
- Adversarial testing: identity stability under prompt injection and deliberate disambiguation attacks
- Extension to collective and organizational identities
- Development of open tooling for ELC deployment accessible to non-technical users

7. Conclusion

Identity in AI-mediated environments is no longer retrieved — it is reconstructed. The fidelity of that reconstruction depends not on visibility but on the architecture of one’s digital presence.

This paper introduced identity entropy as a formal measure of reconstruction divergence and the Entity Life Cycle (ELC) as a practical framework for reducing it. The case study demonstrates that structured, systematic deployment of machine-readable identity artifacts can shift an individual from algorithmic erasure to verifiable entity recognition — without institutional resources.

As generative AI becomes the primary interface through which people discover and evaluate other people, architecting one’s own reconstruction space becomes a fundamental skill. ELC is a first step toward formalizing that practice.

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Related methodology: Native AI Branding · Entity Life Cycle (ELC)

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